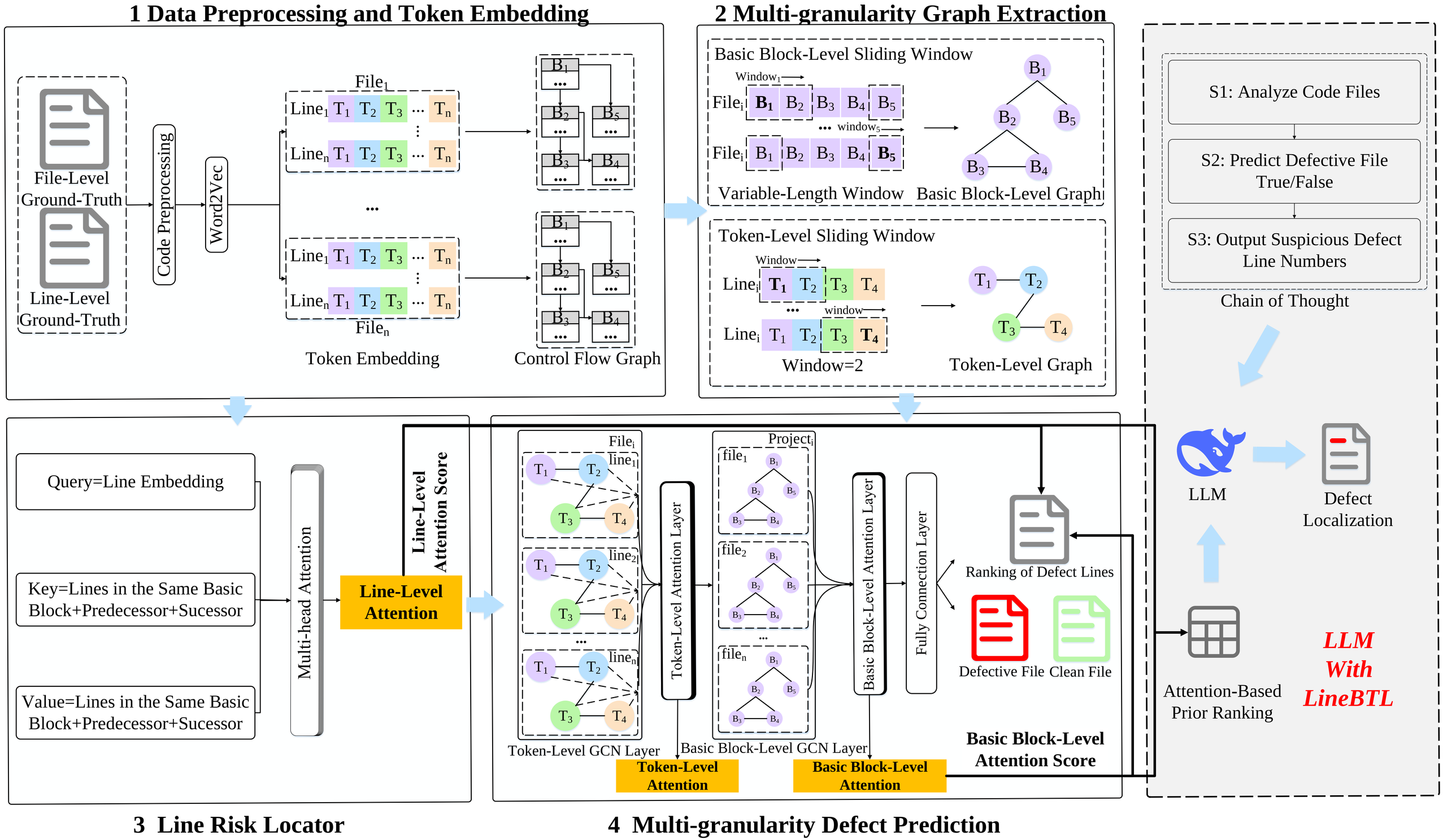
1 Data Preprocessing and Token Embedding
File-Level Ground-Truth
Line-Level Ground-Truth
Code Preprocessing
Word2Vec
File1
Line1 T1 T2 T3 ... Tn
Linen T1 T2 T3 ... Tn
Filen
Token Embedding
Control Flow Graph
2 Multi-granularity Graph Extraction
Basic Block-Level Sliding Window
Window1
Fileᵢ B1 B2 B3 B4 B5
window5
Variable-Length Window
Basic Block-Level Graph
Token-Level Sliding Window
Window
Lineᵢ T1 T2 T3 T4
window
Window=2
Token-Level Graph
S1: Analyze Code Files
S2: Predict Defective File True/False
S3: Output Suspicious Defect Line Numbers
Chain of Thought
Query=Line Embedding
Key=Lines in the Same Basic Block+Predecessor+Sucessor
Value=Lines in the Same Basic Block+Predecessor+Sucessor
Multi-head Attention
Line-Level Attention
Line-Level Attention Score
3 Line Risk Locator
Fileᵢ
line1
line2
linen
Token-Level GCN Layer
Token-Level Attention Layer
Projectᵢ
file1
file2
filen
Basic Block-Level GCN Layer
Basic Block-Level Attention Layer
Fully Connection Layer
Ranking of Defect Lines
Defective File
Clean File
Basic Block-Level Attention
Basic Block-Level Attention Score
4 Multi-granularity Defect Prediction
LLM
Defect Localization
Attention-Based Prior Ranking
LLM With LineBTL